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A Sentiment Analysis System for Demand Forecasting

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Supervisors signatures





Dedication

To my beloved mother Salwa, for her endless love and care and for being the most charming, faithful, forgiving and caring person in the world.

To my father Mohamed, for all the moments of support and encouragement he gave me during my whole life. I promise to achieve all the dreams of your life that you could not do, because you were busy making sure that I achieved mine.

Mum & Dad, thanks for all your sacrifices and for being a great example to follow. To my wonderful brother and sister Yacin & Bochra, for enlightening my life with joy, love and happiness.

To all my family and relatives, to all my dear friends for their support and presence, to all my teachers and professors, I dedicate this modest work as a testimony of my affection and my infinite gratitude.

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My project was proposed by the company Infor, in order to develop and achieve a sentiment analysis system for demand forecasting.

In term of this project, I would especially like to thank my company supervisors ***Mr. Mohamed Ali BEN JEMAA****, who, despite the responsibilities he has assumed, has always found the time to provide me with his valuable pieces of advice. With his precious help in the most delicate moments and his availability, I was able to overcome all the problems I have encountered. May this work be the modest testimony of my high respect.*

It goes without saying that I am grateful to my academic supervisor ***Mrs. Khaoula TBARKI*** *for her remarks and her continuous support. She tried her best to give me her time and effort so that this project comes to surface.*

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General Introduction

The last decade has witnessed a great revolution in both social media and websites for online sales. These two channels have allowed people to voice their opinions and communicate it worldwide. Hence, consumers have any product and/or service reviews available for them before taking any buy decision. This has made online reviews in social media or retailers' websites very important for the retail business.

As retailers main concern is business profitability, improving competitiveness in the market and satisfying customers represent real challenges. Hence, they rely on all available information and indicators to make better decisions. Online reviews constitute a major element of these indicators. Numerous studies [B1] have been carried out to show the impact which reviews have on the popularity, rate of sales, product awareness, conversion rate and profitability of businesses that are selling products or services.

With the growth of Machine Learning discipline, businesses are now able to benefit from all available information. In fact, the latest developments in text mining made possible extracting customers sentiment polarity in an automated fashion. Also, the emergence of attribute-based forecasting models in predicting the future demand made possible incorporating a variety of attributes like sentiment polarity in addition to the traditional time attribute used within the time-series models.

In this context we propose a sentiment analysis system for demand forecasting. In fact, we are building a solution that collects customers reviews from online sources and extract the sentiment polarity of clients towards any given product. We are, also, taking advantage of that information to improve the forecast accuracy using the existing attribute-based forecasting model, this proposed solution is composed of seven steps: data scraping, data cleaning, text vectorization, model selection and forecast improvement. Moreover, our aim consists also in analyzing and showing the

evolution of customers opinion through time, in a way that allows decision makers understand history and take accurate decisions. We go through the same steps that we have seen previously except for improving the forecast accuracy.

Our report is composed of three chapters. In the first chapter, we describe the general frame of the project by introducing the host company, going first through its main highlights, then presenting the project overview. Throughout the second chapter, we introduce our sentiment analysis system, we shed the light on some of the most important keywords of our work while providing a brief literature review. Next, in the third chapter, we evaluate the performance of our solution and discuss the found results. Finally, we conclude our work by giving prospects and potentials performance upgrades.

Chapter I:

General Presentation

I.1 Introduction

This project was hosted in Infor, within the data science service, in order to develop a sentiment analysis system for demand forecasting. This chapter is composed of two principal sections, in the first section, we present the hosting company in which we carried out our project. In second section, we introduce the project frame, the main problem that we want to solve, and the approach that we adopted.

I.2 Hosting Company Presentation

In this section, we present the hosting organization, its business and main activities.

I.2.1 Infor Presentation

Infor is a multi-national enterprise software company, headquartered in New York City, United States. It is the largest privately held technology provider in the world. Infor focuses on business applications for organizations delivered via cloud computing as a service. Infor's Tunis office is part of Infor's Demand Management department which was originally a separate company called **Predictix.**

Predictix, is a consulting and enterprise software company. Located in Atlanta (Georgia, USA), it is a specialized consultancy and software firm with about one hundred employees and a large number of clients around the world. Headquartered in Atlanta, with offices in London and Tunis, Predictix works with different customers worldwide, from the world's largest mass and specialty retailers to up-and-coming fashion retailers in North America, South America, Europe, Asia, Australia and Africa.

Its main goal is to help retailers and retail brands improving their business planning and decision making by implementing and delivering tailored IT solutions. It is a company that specializes in predictive demand management and optimization through applying predictive analytic technologies to solve retail problems.

Predictive Demand or Demand Forecasting is the prediction of customer's future demand based on the history.

Retail is the process of selling consumer goods or services to customers through multiple channels of distribution to earn a profit [N15].

This company has applied and implemented science in various business processes to improve retailers' profitability. Such processes include forecasting, price optimization, inventory management and optimization and planning. Providing the next generation of merchandising solutions, the final product is delivered over the web as a Software as a Service. Predictix/Infor focuses on developing solutions that evolve easily to adapt to the ever-changing client demands.

I.2.3 Data Science Team Presentation

The main task force in Infor's Demand Management department is divided into several teams. The Data Science team's main focus is to provide sales forecast at a granular level which allows afterwards the execution of custom processes like replenishment, assortment planning and supply chain optimization...

Therefore, the science team ought to work with the other teams in order to ensure high quality applications.

I.2.3 Fields of Expertise

The firm provides advice, implementation and software services in the areas of:

- **Forecasting and replenishment:** The firm provides forecasts and replenishment strategies that enable a retailer to implement the promotional pricing and assortment strategies to be established. Additional profit can be gained by minimizing inventory levels and reducing lost sales.
- **Pricing and promotions:** Retailers, wholesalers, and some manufacturers. Infor services to help them manage their promotions, pricing and markdowns for higher profit bottom lines.
- **Assortment and space optimization:** Infor helps retailers defining the optimal localized assortment for every store and every space.
- **Planning and allocation:** Business Consulting at Infor helps retailers review their current planning and allocation processes and design new improved ones

I.3 General Project Presentation

This project aims to develop a new system for sentiment analysis for demand forecasting. In this section, present the problem statement and the project overview.

I.3.1 Problem Statement

Online Reviews have become more influential on customer decision-making due to increasing product complexity and product availability over the internet. Moreover, this type of feedback spreads rapidly among customers and can be accessed anytime and anywhere, which further increases its significance.

As a result, online customer's opinions usually affect important business outcomes such as retail prices, while positive feedback would present an opportunity for companies to make more profit, negative feedback may damage to the prestige, profitability and trustworthiness of the business. Therefore, companies today tend to consider customers feedback in their market analysis.

This project aims to take advantage of the online reviews for retailers. We intend to model their effect on the prediction of products demand. This additional attribute is expected to improve the forecast accuracy when using the attribute-based forecasting models. Moreover, this project also aims to show the importance of analyzing sentiment intensity over time for businesses which indicates the satisfaction and dissatisfaction of customers to help in taking decisions and improving therefore their profitability.

I.2.4 Project Overview

This project has as main goal the design and implementation of a machine learning solution that analyzes online customer's opinions and extracts a sentiment intensity value that ranges between –1 and 1. The values between 0 and 1 indicate positive sentiment, where the negative values indicate the negative sentiment, as shown in the figure I.1.

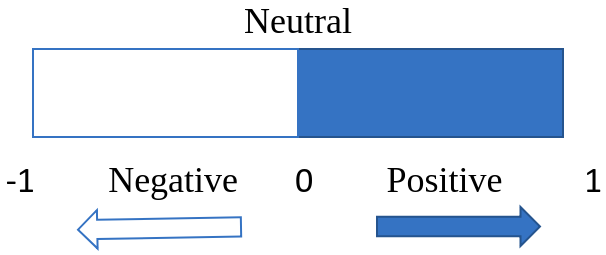


Figure I.1: Sentiment intensity range

We are working on two different data sources, online reviews and tweets; therefore, we apply two different approaches on each one of them for sentiment analysis:

- Sentiment analysis system for demand forecast: The sentiment intensity is returned for each product (based on its related reviews). It is then passed as a feature to Infor's already existing forecasting model, in order to estimate the sales prediction and determine its impact.

- Sentiment analysis system for brand perception: The analysis is based on user's tweets on the brand of one of Infor's retail customers.

I.4 Conclusion

In this chapter, we have presented in general the host company Infor, and the data science team, where this project was carried out. Then, we have described the project's problematic and objectives of our solution. In the next chapter, we show in detail our proposed systems for sentiment analysis.

Chapter II:

Proposed Sentiment Analysis Systems

II.1 Introduction

Sentiment analysis is the process of measuring the inclination of people's opinions through natural language processing (NLP) and text analysis techniques. In other words, we determine the sentiments or reactions towards some products, people or ideas with revealing the contextual polarity of the text [N1].

Sentiment can be categorical, such as, negative, neutral and positive, or it can be numerical like a range of intensities or scores.

This chapter presents a theoretical study of our proposed systems for sentiment analysis, the proposed solutions covers two main goals, sentiment analysis system for demand forecast improvement, where we adopt a solution that uses machine learning techniques, and brand sentiment analysis system through time, where we use a lexicon-based approach. We have two different sections for explaining each approach's architecture.

II.2 Sentiment Analysis System for Demand Forecast

The development of our proposed solution is composed of seven principal phases as shown in figure II.1:

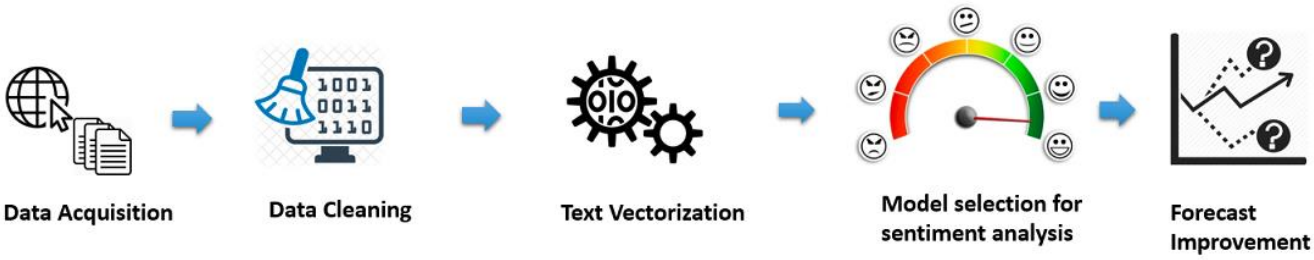


Figure II.1: Design and implementation steps.

- **Data Acquisition**

  The first step is to identify potential sources of data to be used for sentiment analysis and then write web scrapers or use an API (Application Program Interface) code to get the needed data.

- **Data Cleaning and Preprocessing**

  Once the data is collected, it is prepared for analysis by removing or modifying data that is in incorrect, incomplete, irrelevant, duplicated, or improperly formatted.

- **Text Vectorization**

  Text Vectorization consists in using text analysis techniques to create features for machine learning algorithms. It is a very essential step in the project process, since plain text cannot be consumed directly by a machine learning model.

- **Model Selection for Sentiment Analysis**

  In this step, we investigate the existing machine learning models used in sentiment analysis, and choose the one that fits our data. The final output of the model is a sentiment intensity for each product.

- **Forecast Improvement**

  Finally, the estimated sentiment value is passed as a feature to the forecasting model. During this step, determine the impact of the sentiment intensity feature on the existing demand forecasting model

II.2.1 Data Acquisition

This section composed of two sub-sections, data description and data collection.

II.2.1.1 Data Description

As a first step, the goal is to identify potential sources of data that can be used in sentiment analysis system. Therefore, we collect the data about one of Infor's clients, and mainly our focus is on the reviews in its official website.

Customers usually leave reviews on different published products to express their opinions and degree of satisfaction. They also attribute a rating value which ranges between 0 and 5. Figure II.2 shows an example of client's feedback on a product.

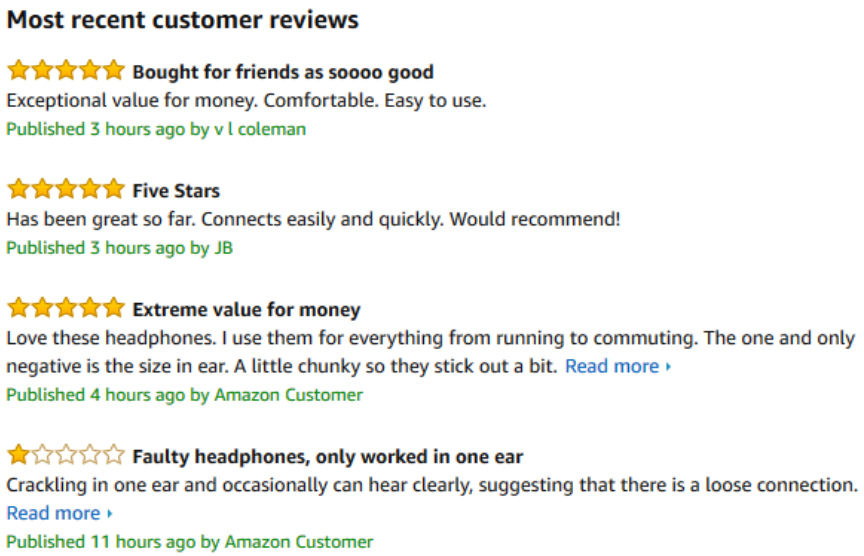


Figure II.2: Example of reviews for online products [N13].

It is important to note that not all products on the website contain associated reviews.

II.2.1.2 Data Collection

For data scraping (collecting data from the web), we use Hypertext Markup Language (HTML) parsing with the help of a scraping framework: Scrapy.

A. HTML Parsing

It is the syntactic analysis of a document, in order to extract structured data using the HTML tag format. This is very used, when we want to scrap data directly from a web page [N2]. HTML is the standard markup language for creating Web pages, a simple HTML document is described in the figure II.3.



Figure II.3: HTML document global architectur [N14].

Where:

-The <!DOCTYPE html> declaration defines this document to be HTML5

-The <html> element is the root element of an HTML page

-The <head> element contains meta information about the document

-The <title> element specifies a title for the document

-The <body> element contains the visible page content

-The <h1> element defines a large heading

-The <p> element defines a paragraph

Parsing HTML consists in locating specific tags that contain the requested reviews. The drawback of this method is that the fact it is always specific to the architecture of the page, it is difficult to use a standard script for scraping multiple webpages.

This technique would be used to get the data from the Infor's client official website. In practice we are going to use Python's crawling framework **Scrapy** to get the data.

B. Scrapy Framework

Scrapy is an open source and collaborative **framework** for extracting from websites. In other words, it is an application framework for crawling web sites and extracting structured data which can be used for a wide range of applications especially in data mining.

Scrapy allows developers to create spiders which are self-contained crawlers that would define how a certain site is going to be scraped. One of its biggest features is that it is built upon an asynchronous networking library, which gives the spider a much greater performance.

A Scrapy project structure includes the following files presented in Figure II.4.

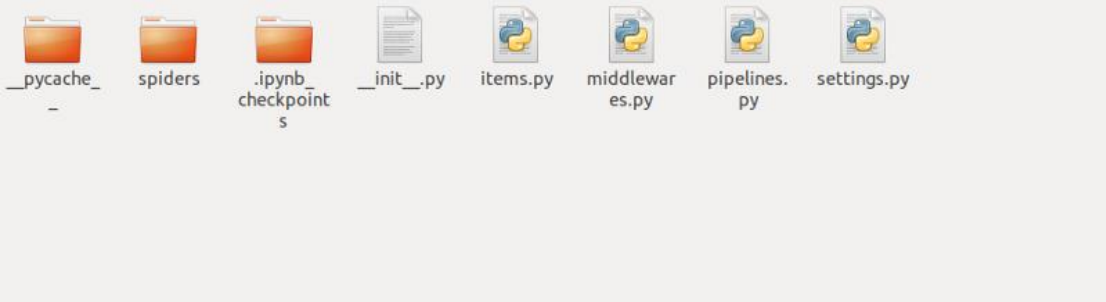


Figure II.4: Scrapy project folder architecture.

where:

- **Items.py:** Items are containers that are loaded with the scraped data.
- **Middleware.py:** The spider middleware is a framework of hooks into Scrapy's spider processing mechanism where you can plug custom functionality to process the responses that are sent to Spiders for processing and to handle the requests and items that are generated from spiders.
- **Pipelines.py:** After an item has been scraped by a spider, it is sent to the Item Pipeline which processes it through several components that are executed sequentially. Each item pipeline component is a Python class.
- **Settings.py:** It allows one to customize the behavior of all Scrapy components, including the core, extensions, pipelines and spiders themselves.

- **Spiders folder:** The Spiders is a directory which contains all spiders/crawlers as Python classes. Whenever one runs/crawls any spider, then Scrapy investigates this directory and tries to find the spider with its name provided by the user. Spiders define how a certain site, or a group of sites are scraped, including how to perform the crawl and how to extract data from their pages.

The data flow in Scrapy is controlled by the execution engine, and goes through the steps presented in figure II.5.

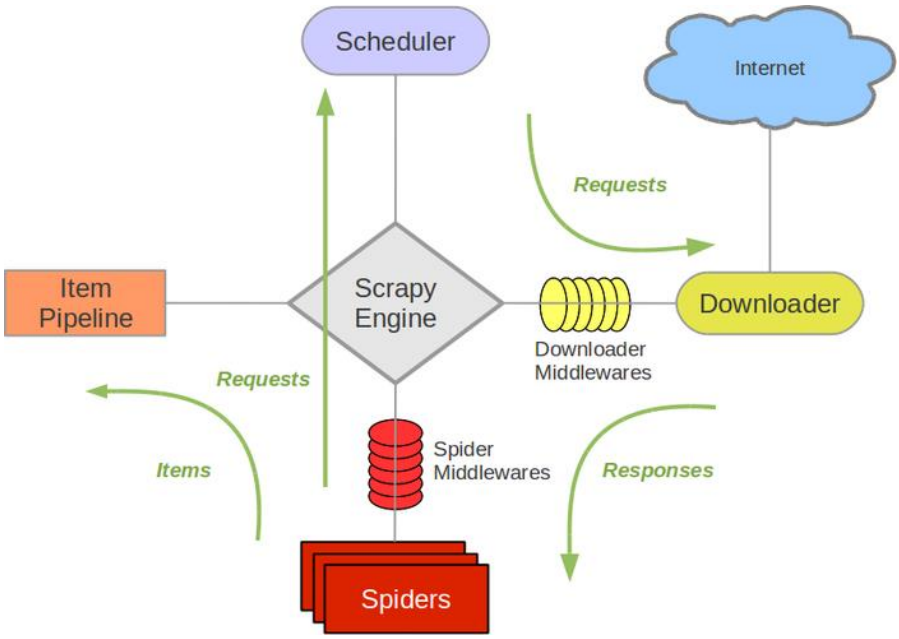


Figure II.5: Data flow through different Scrapy components [N11].

Step 1: The Engine opens a domain, locates the Spider that handles that domain, and asks the spider for the first URLs to crawl.

Step 2: The Engine gets the first URLs to crawl from the Spider and schedules them in the Scheduler, as Requests.

Step 3: The Engine asks the Scheduler for the next URLs to crawl.

Step 4: The Scheduler returns the next URLs to crawl to the Engine and the Engine sends them to the Downloader, passing through the Downloader Middleware.

Step 5: Once the page finishes downloading the Downloader generates a Response (with that page) and sends it to the Engine, passing through the Downloader Middleware.

Step 6: The Engine receives the Response from the Downloader and sends it to the Spider for processing, passing through the Spider Middleware.

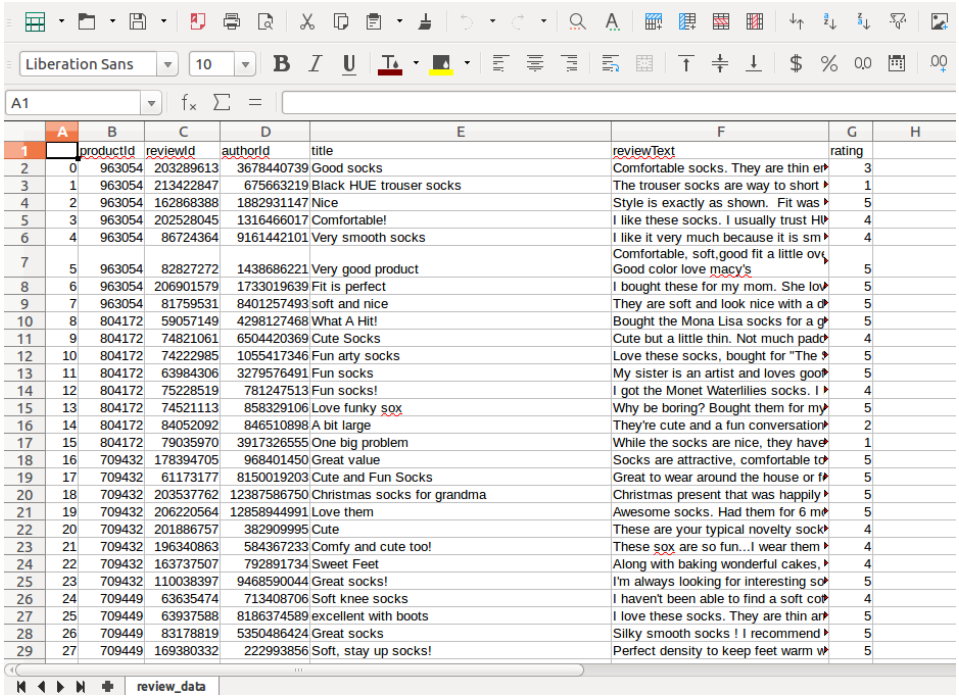
Step 7: The Spider processes the Response and returns scraped Items and new Requests (to follow) to the Engine.

Step 8: The Engine sends scraped Items (returned by the Spider) to the Item Pipeline and Requests (returned by spider) to the Scheduler.

Step 9: The process repeats (from step 2) until there are no more requests from the Scheduler, and the Engine closes the domain.

C. Data Storage

After scraping, the data is stored into files for further processing and analysis, Figure II.6 shows an example of data saved into a Comma Separated Values (csv) file.



	A	B	C	D	E	F	G	H
1		productId	reviewId	authorId	title	reviewText	rating	
2	0	963054	203289613	3678440739	Good socks	Comfortable socks. They are thin er	3	
3	1	963054	213422847	675663219	Black HUE trouser socks	The trouser socks are way to short	1	
4	2	963054	162868388	1882931147	Nice	Style is exactly as shown. Fit was	5	
5	3	963054	202528045	1316466017	Comfortable!	I like these socks. I usually trust HU	4	
6	4	963054	86724364	9161442101	Very smooth socks	I like it very much because it is sm	4	
7	5	963054	82827272	1438686221	Very good product	Comfortable, soft,good fit a little ove Good color love macy's	5	
8	6	963054	206901579	1733019639	Fit is perfect	I bought these for my mom. She lov	5	
9	7	963054	81759531	8401257493	soft and nice	They are soft and look nice with a d	5	
10	8	804172	59057149	4298127468	What A Hit!	Bought the Mona Lisa socks for a g	5	
11	9	804172	74821061	6504420369	Cute Socks	Cute but a little thin. Not much pad	4	
12	10	804172	74222985	1055417346	Fun arty socks	Love these socks, bought for "The	5	
13	11	804172	63984306	3279576491	Fun socks	My sister is an artist and loves goo	5	
14	12	804172	75228519	781247513	Fun socks!	I got the Monet Waterlilies socks. I	4	
15	13	804172	74521113	858329106	Love funky sox	Why be boring? Bought them for my	5	
16	14	804172	84052092	846510898	A bit large	They're cute and a fun conversation	2	
17	15	804172	79035970	3917326555	One big problem	While the socks are nice, they have	1	
18	16	709432	178394705	968401450	Great value	Socks are attractive, comfortable to	5	
19	17	709432	61173177	8150019203	Cute and Fun Socks	Great to wear around the house or f	5	
20	18	709432	203537762	12387586750	Christmas socks for grandma	Christmas present that was happily	5	
21	19	709432	206220564	12858944991	Love them	Awesome socks. Had them for 6 mo	5	
22	20	709432	201886757	382909995	Cute	These are your typical novelty sock	4	
23	21	709432	196340863	584367233	Comfy and cute too!	These sox are so fun...I wear them	4	
24	22	709432	163737507	792891734	Sweet Feet	Along with baking wonderful cakes,	4	
25	23	709432	110038397	9468590044	Great socks!	I'm always looking for interesting so	5	
26	24	709449	63635474	713408706	Soft knee socks	I haven't been able to find a soft cot	4	
27	25	709449	63937588	8186374589	excellent with boots	I love these socks. They are thin an	5	
28	26	709449	83178819	5350486424	Great socks	Silky smooth socks ! I recommend	5	
29	27	709449	169380332	222993856	Soft, stay up socks!	Perfect density to keep feet warm w	5	

Figure II.6: Overview of the scraped data in a CSV file.

II.2.2 Data Cleaning

Data preparation is an important phase before applying any machine learning algorithm. The data preparation is done by data preprocessing. The preprocessing of text means cleaning of noise, such as, stop words, punctuation, terms, which don't carry much weightage in context to the text.

The initial steps are:

- Deleting null values
- Deleting duplicates
- Deleting tabs
- Deleting bad symbols and punctuations

Once the data is well cleaned, we go through the following process:

- **Tokenization:** It is dividing text into small pieces (tokens) that are easier to process, generally we use space as a splitting delimiter for English language which won't be the case for other languages like for example German.

- **Stop words removal:** Stop words are the words that are commonly used in the language (such as "the"), they usually don't carry any sentiment expressed by the reviewer that's why they are removed.

- **Stemming/lemmatization:** Both terms are used to generate the root form out of a specific word, The difference is that **stem** might not be an actual word whereas **lemma** is an actual language word derived from the dictionary. For example the stem of beautiful and beauty is "beauti" which has no meaning in English dictionary by contrast to the lemma which would be "beautiful".

II.2.3 Text Vectorization

The inherent unstructured and noisy nature of textual data makes it harder for machine learning methods to directly work on raw text data. From here comes the need of transforming sentences into numerical feature vectors that can be processed using machine learning models.

There are multiple machine learning solutions that could be used for the text vectorization step. Among them, we can mention: N-gram Model, Term Frequency–Inverse Document Frequency and Word Embeddings.

- N-gram Model

It consists of building a feature vector containing all the tokens within the training corpus. A token could be one word or an n-gram, which is a contiguous sequence of n items (n=1 for a unigram, n=2 for bigram, n=3 for trigram...). From a given sample of text, different n-gram sizes may be used to model contextual features within the text. Each training sample is then represented using a vector of different tokens frequencies that are contained within it.

- Term Frequency–Inverse Document Frequency

TF-IDF (short for term frequency–inverse document frequency) is intended to reflect how important a word is to a text sample in a collection or corpus, Term frequency is the raw count of a term in the text, while Inverse document frequency is the logarithmically scaled inverse fraction of the other text samples that contain the word.
TF-IDF values are used together with the n-gram model, the difference is that we put the product between TF and IDF as a feature instead of term frequency:

$$w_{ij} = tf_{ij} \times \log_2 \frac{N}{n} \qquad \text{II.1}$$

where, w_{ij} is the weight of term T_j in document D_i, tf_{ij} is the frequency of term T_j in document D_i, N is the number of documents in collection and n is the number of documents where the T_j occurs at least once. A document is a training text sample.

- Word Embeddings

It is a representation of document vocabulary while capturing the context words, semantics, syntactic similarity and relation with each other. Such a, representation could be obtained using two main methods: Continuous Bag of words Model and Skip – Gram model.

Continuous Bag of words Model: (CBOW) this method takes the context of each word as the input and predicts the word corresponding to the context, the corresponding architecture is as follows in Figure II.7.

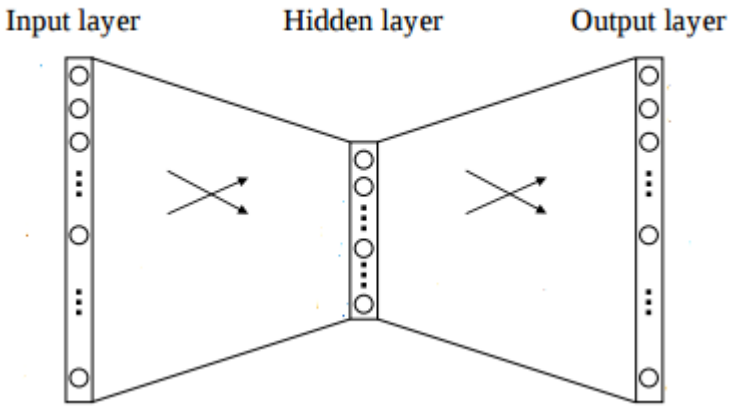


Figure II.7: Continuous bag of words architecture [N4].

The Input layer is an n-gram representation of a word with the use of 0 or1 flag presence instead of term frequency for the feature's values.

After training the neural network, the values of the Hidden layer serve as the representation of a word in function of its context.

The Output Layer has the same form as the input layer except it is for the neighbor of the given word.

For words that have multiple contexts, the result is the average of all of them.

Skip – Gram model: The aim of skip-gram, as opposite to CBOW, is to predict the context given a word, thus for each word context there is a specific target value and output (for one input word we have multiple output context words), as it is described in Figure II.8.

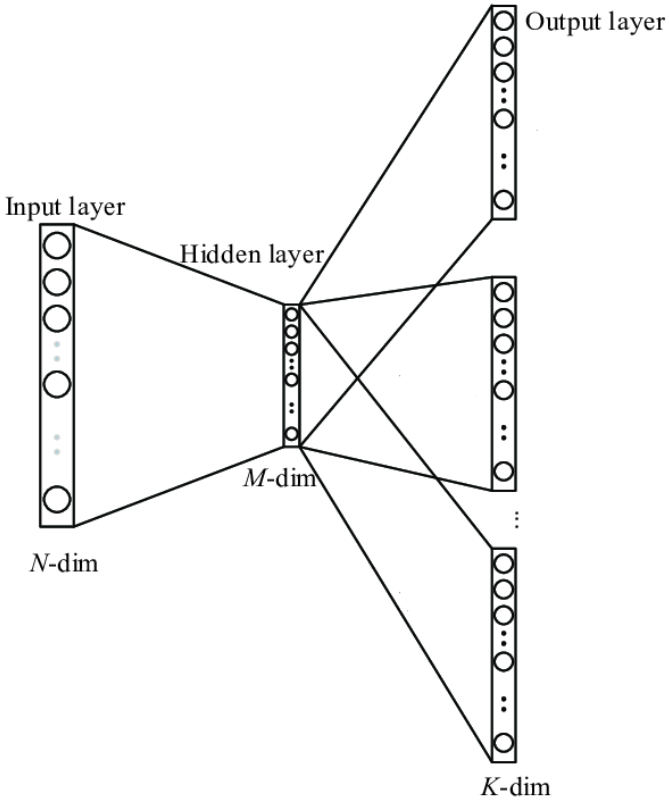


Figure II.8: Skip-Gram model architecture [N5].

As shown in the Figure II.8, there is different outputs for one single word depending on its different contexts.

In term of comparison between mentioned solutions above, we can remark that TF-IDF and Word Embeddings can give better results than the n-gram model. TF-IDF has the advantage of extracting important words in text, while Word Embeddings helps in modeling the context.

II.2.4 Model Selection

This step presents different algorithms and techniques, in order to choose the appropriate algorithm, which gives better results for sentiment analysis in our case.

Machine learning approach is constituted of two main steps, as shown in figure below (Figure II.9).

- **Training step**

  This step allows to build a classification model using training data. In our case, we have two classes positive and negative customer's opinion, so we need data of both classes and their labels, as we have a supervised classification problem.

- **Testing step**

  Based on the obtained classification problem, we test it using unseen data that contain both classes. And we evaluate the model performance using some experimental protocols, to choose the optimal parameter.

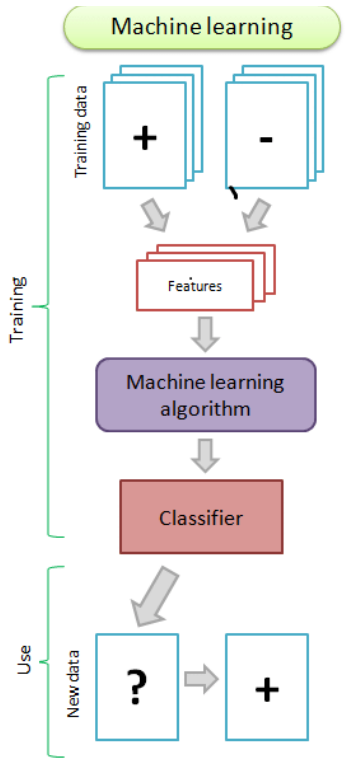


Figure II.9: Machine learning approach architecture [N3].

The table II.1 presents a list of the well-known machine learning models that we can choose for our sentiment analysis system together with their advantages and limits.

Table II.1: Comparison between machine learning models for sentiment analysis

Model	Core idea	Advantages	Drawbacks
Naive Bayes	Given a data sample, this model would return the class with the highest probability, it makes the assumption of feature's independence between each other.	It needs less training data then other models. It is not sensitive to irrelevant features.	It makes a very strong assumption on the shape of the data distribution,
Maximum Entropy	It is a probabilistic classifier that forms all the models that fit our data, and then selects the one which has the largest entropy (most informative).	It is used even when we don't know anything about the prior distributions of data.	It requires a lot of training time due to the optimization problem that needs to be solved in order to estimate the parameters of the model.
Support Vector Machine (SVM)	SVM is a large-margin classifier formally defined by a separating hyperplane that categorizes different samples to their corresponding classes.	Using SVM with combination of the kernel method (defined I chapter two) would make complex nonlinear decision boundaries possible for the separation between classes.	It finds difficulties learning structured representations of text because it does not model order information within it.
Convolutional Neural Network (CNN)	CNN is an artificial neural network which utilizes layers with convolving filters that are applied to training data in order to extract a useful pattern in the features for final data classification.	Considers local structure since a convolution is usually taken over adjacent terms, information contained in adjacent words is learned effectively.	In sentences, words often don't need to be adjacent to be related. This makes CNN perform badly for modeling the total sequence of words (global architecture)

Model	Core Idea	Advantages	Drawbacks
Recurrent Neural Network (RNN)	RNN is an artificial neural network where connections between nodes form a directed graph along a temporal sequence. (as shown in figure number in chapter two)	Given that text is naturally sequential, RNN works best for sentiment analysis. It can be used with a Long Short-Term Memory (LSTM) extension for higher memory capacities	It is difficult to train and requires a lot of running time.

Among the described models, we picked SVM and LSTM for testing and evaluation, the next two paragraphs explain the algorithms and the motivation behind choosing them:

- Support Vector Machine

  Definition

  Support Vector Machine, short for SVM is a large-margin classifier formally defined by a separating hyperplane. In other words, given labeled training data, the algorithm outputs an optimal hyperplane which categorizes all the samples as shown in Figure II.10.

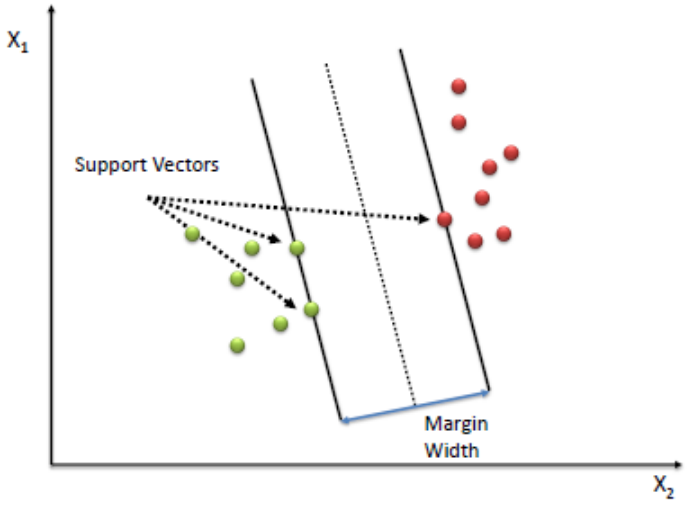


Figure II.10: Separation between classes using support vector machine [N6].

The points that are close to the hyperplane are called support vectors and thy are used for determining the separation boundary between the classes. Optimal hyperplane is obtained by solving the following equation:

minimize:

$$W(\alpha) = -\sum_{i=1}^{l} \alpha_i + \frac{1}{2}\sum_{i=1}^{l} \quad . \sum \quad y_i y_j \alpha_i \alpha_j x_i x_j \qquad \text{II.2}$$

Subject to:

$$\sum_{\substack{i=1 \\ 0 \le \alpha_i \le c}}^{l} y_i \alpha_i = 0$$

where l is the number of data points in our training data
x is the feature vector in each training example
y denotes the outputs of the data points
alpha is the Lagrangian constant used for finding the local minima.

Sometimes it is impossible to find a single line that separates the classes, that's why a kernel could be used to project the data in to a higher dimension from the original space, this would allow us to draw non-linear hyperplanes as shown for example in Figure II.11.

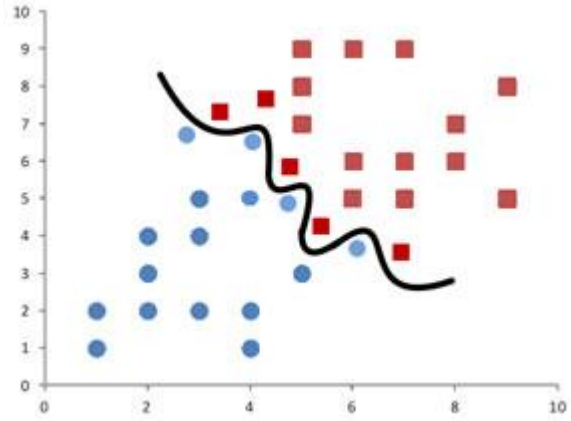


Figure II.11: Separation between data using a nonlinear hyperplane [N7].

It is possible to use SVM with regularization option in order to minimize overfitting, Overfitting happens when a model learns the detail and noise in the training data to the extent that it negatively impacts the performance of the model on new data, regularization consists of putting a constraint on the model's weights, that penalizes high learned values.

Motivation behind Choosing SVM

It is capable of drawing complex decision boundaries and giving high accuracy for separation between classes compared to the other machine learning models, besides, it can be applied to any kind of vectors which encode any kind of data, so it would work well with text vectorization methods.

- Long-Short Term Memory

Definition

A recurrent neural network (RNN) is an artificial neural network where connections between nodes form a directed graph along a temporal sequence. This allows it to exhibit temporal dynamic behavior of the total data, its neurons contain loops so that information persists.

Long Short-Term Memory (LSTM) networks are an extension for recurrent neural networks, which basically extends their memory. Therefore, it is well suited to learn from important experiences that have very long-time lags in between.

The units of an LSTM are used as building units for the layers of the RNN, which is then often called an LSTM network as shown in Figure II.12.

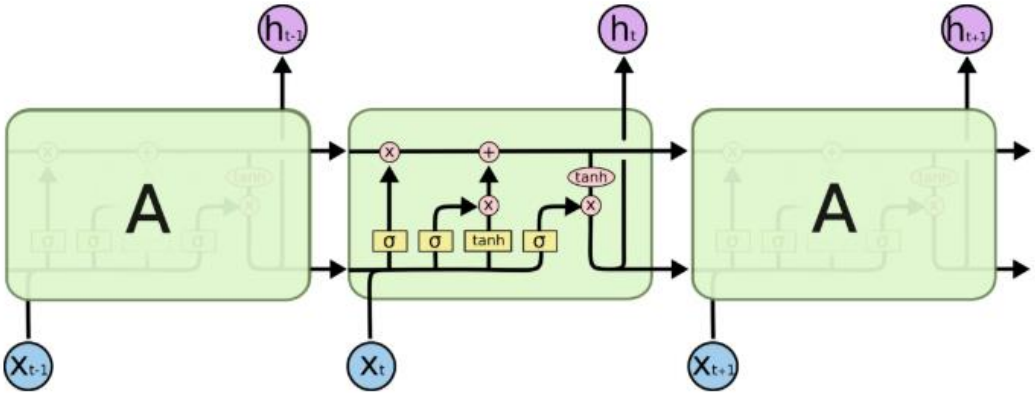


Figure II.12: LSTM units architecture [N8].

LSTM's enable RNN's to remember their inputs over a long period of time. This is because LSTM's contain their information in a memory, that is much like the memory of a computer because the LSTM can read, write and delete information from its memory.

This memory can be seen as a gated cell in the form of sigmoids that range from 0 to 1, where gated means that the cell decides whether or not to store or delete information (if it opens the gates or not), based on the importance it assigns to the information. The assigning of importance happens through weights, which are also learned by the algorithm. This simply means that it learns over time which information is important and which not. Here is a closer look at an LSTM unit architecture in Figure II.13.

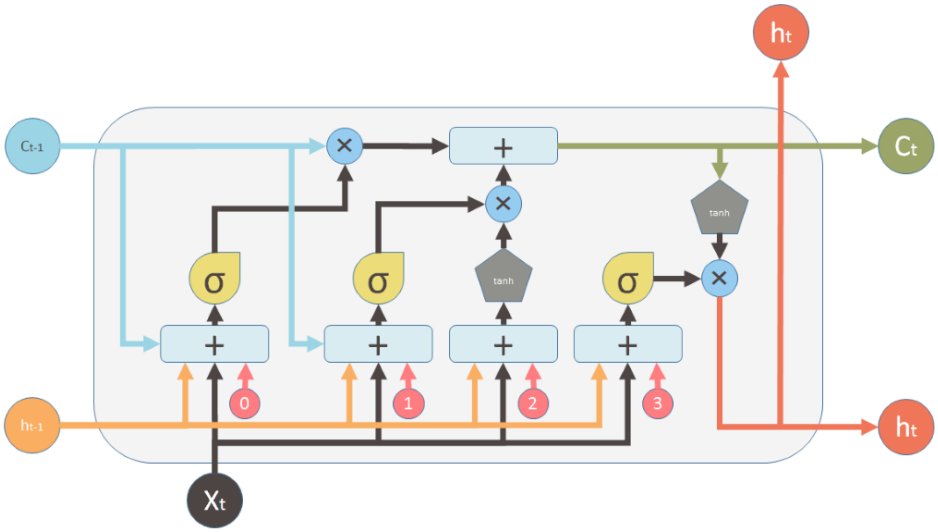


Figure II.13: Different components of an LSTM unit [N9].

Where:

X_t is the input vector	C_t is the memory from previous block
h_{t-1}: is the output of previous block	C_{t-1} is the memory from current block
h_t: is the output of current block	∂ is the sigmoid function
Tanh: is the hyperbolic tangent	X is element-wise multiplication
+ is element-wise addition	

Motivation behind Choosing RNN (LSTM)

Being constituted of nodes that are connected along a temporal sequence, RNN is likely to show good results when dealing with textual data, in addition an LSTM unit could be useful for remembering words far neighbors.

The already mentioned models are used for sentiment classification (positive/negative) the final output is expected to be a sentiment intensity that ranges between –1 and 1, Therefore, we take the probability of the predicted class as our numerical output. The approach that we adopted is in Table II.2.

Table II.2: Adopted approach for sentiment intensity estimation.

Predicted class is 1 (positive)	**Predicted class is 0 (negative)**
We take the probability of the predicted class as it is.	We take **minus** the probability of the predicted class.

The models return a value between 0.5 and 1 for the predicted class, we use scaling techniques to transform the range back into between 0 and 1.

II.2.5 Forecast Improvement

II.2.5.1 Infor Forecasting solution

Infor is using a machine learning solution for her retailers' client's sales prediction based on the **Factorization Machine** model, it is a general-purpose supervised learning algorithm that can be

used for both classification and regression tasks, it is an extension of a linear model that is designed to capture interactions between features within high dimensional sparse datasets economically.

FaMa (short for Factorization Machine) is an attribute-based forecasting model in fact, it tries to capture the interactions between different demand drivers and evaluate their impact on the forecast, FaMa uses attributes (features) to model the sales demand drivers, the definition of an attribute or a feature is the properties of a problem based on which we would like to predict results.

II.2.5.2 Approach

Sentiment analysis is thought to be a product's demand driver, therefore, it is passed as a feature to the FaMa model, we then run the sales forecast with and without the sentiment feature for one of Infor's client and analyze the results.

The process that we have gone through is resumed in Figure II.14.

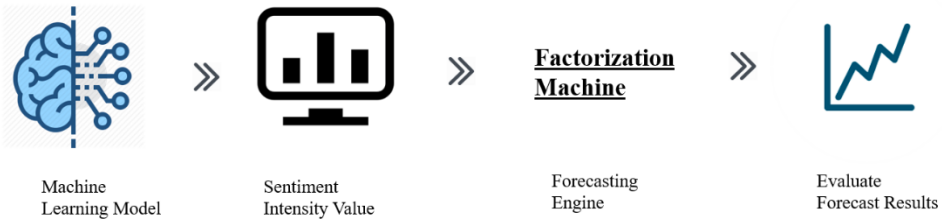


Figure II.14: Forecast improvement process.

II.3 Sentiment Analysis System for Brand Perception

During this section, we focus on analyzing tweets of users about one of Infor's clients, and we aim to extract the evolution of customer's opinion through time

II.3.1 Data Acquisition

Our source of data is twitter where we search for tweets that mention our target company:

- **Twitter:**

Twitter is one of the most popular social networking sites where people are expressing their views, opinions and emotions freely. These tweets could be recorded and analyzed to mine customers opinion about the brand and its products. Though it will not be possible to extract tweets per product identifier, therefore, we analyze the global attitude of the customers towards the whole brand and see the different variations of this attitude through time. Figure II.1 represents an example of a client's complaint on Twitter.



Figure II.15: User's complaint on Twitter [N10].

We use an API call to scrape the tweets from the internet:

- **API Definition:**

An API (Application Programming Interface) is the way computer programs "talk" to each other so that they can request and deliver information. This is done by allowing a software application to call what's known as an **endpoint**: an address that corresponds with a specific type of information we provide (endpoints are generally unique like phone numbers).

- **Twitter API:**

Twitter allows access to parts of his service via APIs to allow people to build software that integrates with Twitter, like a solution that helps a company respond to customer feedback on Twitter similar to our case.

Authentication: All requests to the Usage API require HTTP (Hypertext Transfer Protocol) Basic Authentication, using any of the email/password credentials enabled on the user's account. Twitter API is public and so it offers the possibility to retrieve other user's tweets.

Usage: The following API endpoints shown in Table II.3 can be used to programmatically create, retrieve and delete Tweets, Retweets and Likes.

Table II.3: Different twitter endpoints.

Tweets	Retweets	Likes (formerly favorites)
• POST statuses/update • POST statuses/destroy/:id • GET statuses/show/:id • GET statuses/oembed • GET statuses/lookup	• POST statuses/retweet/:id • POST statuses/unretweet/:id • GET statuses/retweets/:id • GET statuses/retweets_of_me • GET statuses/retweeters/ids	• POST favorites/create/:id • POST favorites/destroy/:id • GET favorites/list

For our case, we are interested in GET API used for retrieving tweets, precisely we will be looking for tweets that are related to Infor's client since he is our study case.

The scraped tweets are stored in a csv file similar to online reviews.

II.3.2 Data Cleaning

In the case of tweets, the data is very noisy and needs special care defined by initial cleaning steps which could be summarized in Table II.4.

Table II.4: Different cleaning tweet's steps.

Cleaning step	Operation	Explanation
Twitter @user tag	Convert to 'USER' tag	This tag does not carry any sentiment, it just specifies a user's name
Twitter hashtags #tag	Remove the hashtag symbol, the word remains	The hashtags usually carry a sentiment that is expressed by the user
Non-English tweets	Dropped	They represent only a small portion of the total data, available free API's for translation put constraints on the amount of translated tweets.
Hyperlink	Convert to 'LINK' tag	It does not contain any opinion of the user, so the 'LINK' tag is used to conserve the context of its presence in the sentence. In case of tweets that contain only a hyperlink they are deleted
Numbers	Convert to 'NUMBER' tag	A 'NUMBER' tag is kept for conserving the context of sentences

As for emoticons, each group of symbols is converted to its equivalent sentiment label, Table II.5 gives the mapping used between symbols and emotions.

Table II.5: Emoticons list with their equivalent meanings.

Emoticon	Tag
:), :-), (:, (-:, :')	Smile
:D, :-D,xD, XD	Laugh
;), ;-), (;, (-;, ;-D	Wink
<3, :*	Love
:(, >:(, :-(,):,)-:	Sad
:, (, :'(, :"(	Cry

The different patterns are matched using regular expressions which allows an automated detection process. Here is an example of tweets before and after the cleaning process in Figure II.16 and Figure II.17.

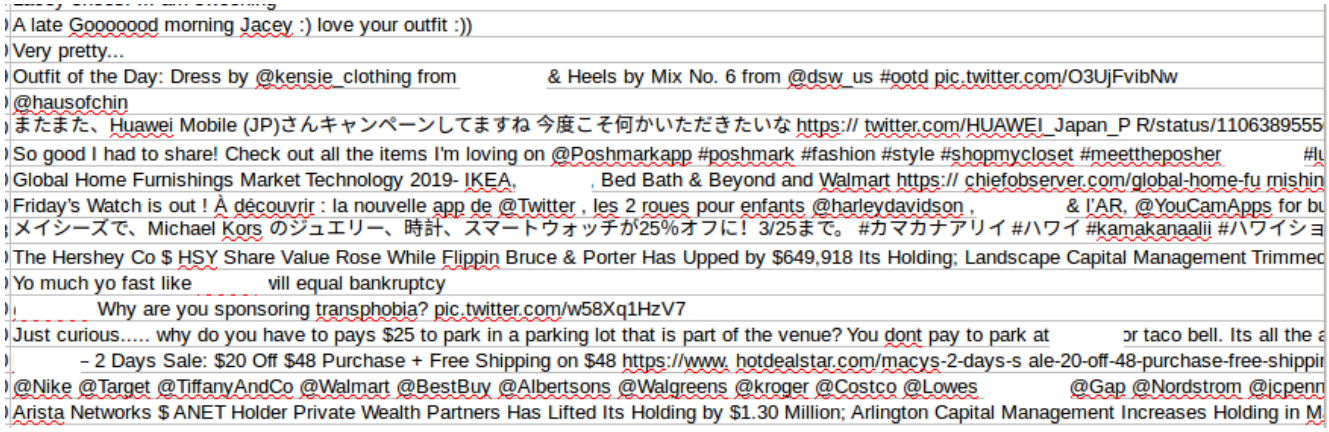


Figure II.16: An overview of tweets before cleaning.

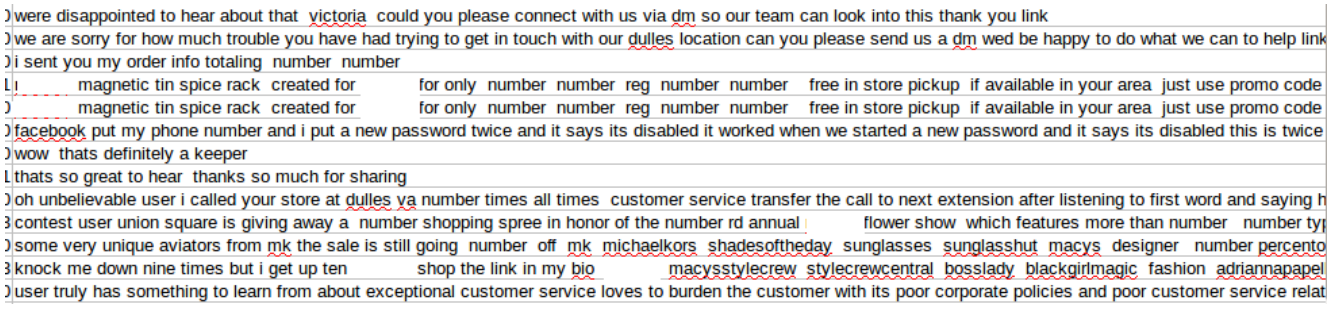


Figure II.17: An overview of tweets after cleaning.

II.2.3 Model Selection

In this step, we propose a sentiment analysis system for customer's opinion in tweets.

Tweet's sentiment analysis is different than online reviews, it is impossible to identify associated tweets for each product, sometimes a user is just complaining about the service in global and no product in particular, as previously shown in Figure II.1. Therefore, we analyze the evolution of customers attitude through time towards the whole brand, and we output is a sentiment intensity that varies through time. This value translates how the customers are feeling about the brand.

Another characteristic of tweets sentiment analysis is that they do not have labels, our proposed solution consists in using a lexicon-based approach for sentiment intensity estimation:

The sentiment of the sentence is classified based on the kind of words used in it. This approach involves a lexical database and some rules based on the grammatical structure of the language for the sentiment estimation. A lexicon, by definition, is a dictionary that builds a mapping between words and root sentiment meanings (whether the word is considered as good, bad, disappointing…). The method allows to quickly and accurately recognize the text sentiment [B2].

To determine the sentiment polarity of text we go through two main steps:

- Development Phase: Training data is labeled manually using human raters with values that symbolize the sentiment intensity of the text, then we try to evaluate the impact of syntactical and grammatical rules together with the used words on the attributed labels.

- Usage: The extracted lexicons and rules are then used on new data to estimate its corresponding sentiment.

Figure II.18 summarizes the process.

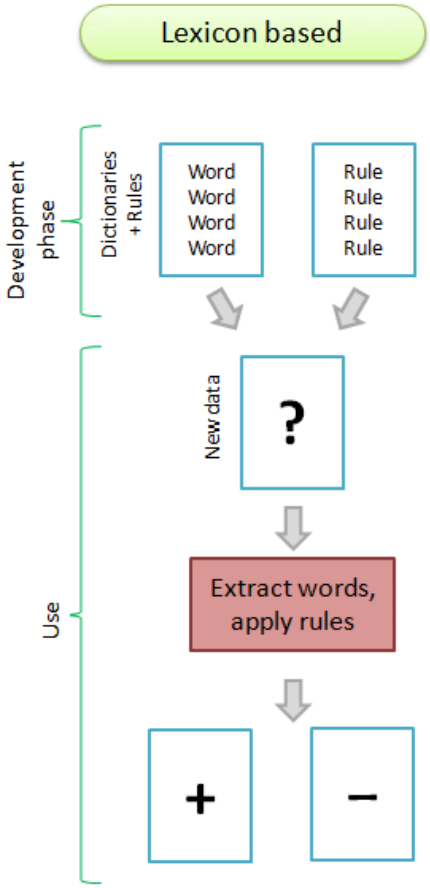


Figure II.18: Lexicon-based approach architecture[N12] .

We choose **Valence Aware Dictionary for sEntiment Reasoning (VADER)** model for tweets sentiment analysis, the model produces and then empirically validates a group of sentiment lexicon together with grammatical and syntactical conventions that humans use when expressing a sentiment.

The training data for the model is labeled using human raters, then we apply a set of qualitative and quantitative methods to extract a set of lexicons and rules. A lexicon, by definition, is a dictionary that builds a mapping between words and root sentiment meanings (whether the word is considered as good, bad, disappointing…), we attribute different lexicons and rules for each class.

Once the training finishes, we use extracted results for sentiment intensity estimation, in fact, VADER algorithm outputs sentiment scores for 4 classes of sentiments that range between –1 and 1:

- Neg: The negativity score of the text
- Pos: The positivity score
- Neu: The neutrality score
- Compound: It is an aggregated score between all the sentiment classes, this value is used as a global indicator of the text sentiment polarity (1 for positive, 0 for neutral, -1 for negative)

VADER model is tightly dependent to domain and language, it was developed on social media content that is written in English. Thus, the established lexicons and rules make a good chemistry with tweets data.

VADER's rules and lexicons were established in a supervised learning manner, thus it is expected to outperform unsupervised learning methods for sentiment analysis. The model's rules and lexicons are already established, so VADER can be used on unlabeled data like tweets.

Chapter III:

Sentiment Analysis Systems Implementation

III.1 Introduction

We detail in this chapter the implementation of our two sentiment analysis systems, for tweets and for nine reviews. We start with describing the work environment, then, we go through the implementation steps for each system.

III.2 Setup of the Work Environment

In this section, we detail the specifications of the machines as well as different software and technologies used in this work.

III.2.1 Hardware

During the process of implementing our project, we have used a local machine, and a distant server. Following are the specifications of these machines:

Local Machine

- CPU: Intel(R) Core(TM) i5-3230M CPU @ 2.60GHz
- Memory: 16,00 GB
- Disk: 500 GB

Server

- CPU: Intel(R) Xeon(R) CPU E5-2686 v4 @ 2.30GHz
- Memory: 32,00 GB
- Disk: 1 TB

III.2.2 Software

To implement the models that we have conceived, we used a wide range of technologies and software that we list here:

Python

Python is a high-level programming language designed to be easy to read and simple to implement. It is open source, which means it is free to use, even for commercial applications. Python can run on Mac, Windows and Unix systems and has also been ported by Java and .NET virtual

machines. It comes with several very attractive machine learning libraries as well as powerful set of packages for analytical needs. This has encouraged us to choose python over the other programming languages. Furthermore, Python has a very large community that provides support through forums, etc.

Numpy

Numpy is the fundamental package for scientific computing with Python. Numpy allows us to process arrays and perform operations related to linear algebra.

Pandas

Pandas is an open source library providing high-performance, easy-to-use data structures and data analysis tools for the python programming language.

Tensorflow

Tensorflow is an open source software library for high performance numerical computation. Its flexible architecture allows easy deployment of computation across a variety of platforms. Originally developed by researchers and engineers from the Google Brain team, tt comes with strong support for machine learning and deep learning and the flexible numerical computation core is used across many other scientific domains.

Keras

Keras is a high-level neural networks API, written in Python and capable of running on top of TensorFlow. It was developed with a focus on enabling fast experimentations with deep learning. Being able to go from idea to result with the least possible delay is key to doing good research. Keras contains numerous implementations of commonly used Artificial Neural Network building blocks such as layers, activation functions, optimizers, etc.

Scikit-learn

Scikit-learn is one of the best-known libraries which provide solid implementation of machine learning algorithms. It contains simple and efficient tools for data mining and data analysis.

Scrapy

Scrapy is python's framework for html parsing and data scraping from the web. It uses specific python classes that are called spiders to extract the data from the HTML web pages.

Twitter API

It is the API offered by the twitter community to access and send data such as messages and media without having to open the Twitter application.

III.3 Implementation

For each one of our proposed sentiment analysis systems we use a different dataset, while online reviews are specific to each product, tweets are about the company's brand in general. We explain in this section, the two used datasets, the error metrics used for performance evaluation, and the results for our two systems.

III.3.1 Used Datasets

III.3.1.1 Tweets

For tweets the collected data is around 60 thousand tweets dating from 2018-09-08 to 2019-03-25 which is about six months' time period. The dataset comprises the following headers:

- Username (string): Name of the user
- Date (string): date of publishment
- Retweet (integer): Number of times the tweet has been shared
- Favorites: Number of times the tweet has been marked as favorite
- Text (string): The textual content of the review
- Hashtags (string): Existing hashtags within the tweet
- Id (string): An identifier for each tweet
- Permalink: A permanent link for accessing the tweet

Figure III.1 represents an overview of the scraped tweets.

	username	date	retweets	favorites	text
0	Fab_Simpsons	2019-03-25 09:27	0	0	Lose yourself in a realm of boundless beauty— @Macys Flower Show is a space-age paradise you've got to see! https:// mcys.co/2n7jPrQ
1	yuafce	2019-03-25 08:35	0	0	$25 / $50 / $100 / $250 / $500 Macys Physical Gift Card - FREE 1st Class Mail (195 Watch count) http:// rover.ebay.com/rover/1/711-53 200-
2	hipshopdeals	2019-03-25 08:13	0	0	Macys - 40% off Pendelton Use code: No Code Needed Ends: 3/26/2019 https:// hipshopdeals.com/macys-40-off-p endelton/ ... https:// hipsh
3	BetterBetsNY	2019-03-25 08:10	0	2	Macy's Flower Show opened at Herald Square. #macys #macysflowershow #macys34th #heraldsquare #34thstreet #sunday #space #journeyt
4	rozycamacho	2019-03-25 08:07	0	0	Come into Macys and find the perfect Juicy fragrance for you! We have a Juicy fragrance for anyone. Or browse form the many selections we
5	paolettacapo	2019-03-25 08:01	0	0	via #BoF - Looking for your next job opportunity in New York? Top brands including RalphLauren, Macys and more are hiring: https:// bit.ly/2KU
6	CCPLUSMBMEM	2019-03-25 07:54	0	0	The #design industry normalises their flagrant denial, all over the internet. Sales @macys for component comparison hikes the sales tax caus
7	MoviesEarly	2019-03-25 06:54	0	0	pic.twitter.com/5vc2YzhOo7
8	ScottKMT1111	2019-03-25 06:50	0	0	I will say, besides the flawless and awesome reporting, the entire look you had going tonight was amazing! Loved the hair, the make-up and ev
9	la_jeannette2	2019-03-25 06:17	0	3	Wow, you are beautiful, loving and smart Shout out to the high school who gave the children who are special have the opportunity to model pro
10	Macys	2019-03-25 06:07	0	0	Please allow our team to look into this for you. Can you please send us a Direct Message with additional details at the link below? Thank you!
11	kikitakesnokaka	2019-03-25 06:02	0	0	@Macys placed an order online but it says "inventory system is offline and unable to verify stock" why take the order at all?
12	iPhone__Fashion	2019-03-25 05:40	0	0	Converse Men's Chuck Taylor Street Mid Sneakers + $25 Macys eGC 2 for $52.50 (after Slickdeals Rebate) + Free Store Pickup http:// iphon
13	Kahnaynay1969	2019-03-25 05:26	0	0	Oh they are gorgeous!!!!
14	EricEtienne0911	2019-03-25 05:22	0	0	@Macys @AventuraMall pic.twitter.com/3MECfkrXjl
15	Vision_I_See	2019-03-25 05:19	0	0	Thank You
16	Amoney8dirty1	2019-03-25 05:16	0	0	So today I went out to macys today and bought a queen size bed sheet set and 4 new pillows lol
17	ebuyerstore	2019-03-25 04:50	0	0	Converse Men's Chuck Taylor Street Mid Sneakers + $25 Macys eGC 2 for $52.50 (after Slickdeals Rebate) + Free Store Pickup http:// ebuye
18	kaylavarggas	2019-03-25 04:45	0	1	someone charged 1900$ to my moms macys account and 1000$ to her ulta account how the fuck yall just be stealing money like its nothing

Figure III.1: Scraped tweets with their headers.

III.3.1.2 Online Reviews

For online reviews, a major part of the products on the website do not have any customer's feedback, therefore the scraped data is not enough to train a machine learning model.

The solution is to use an online free dataset for training and to use the scraped data for final testing. Amazon Reviews Dataset which contains reviews about different products sold on Amazon is chosen. It is an open dataset that is available for public usage. The products are chosen from different categories on purpose so that the final result of our machine learning model is general no matter what the context of the analyzed text is. In fact, this would allow Infor to use the presented solution in multiple projects with different backgrounds.

The dataset contains three different columns:

- Review Title: Title of the review
- Review Text: Usually a long free-style text of the customer point of view regarding the product
- Rating: An integer between 1 and 5 that is attributed by the user to indicate his degree of satisfaction with the product.

The mapping between the ratings and their meanings is the following:

1 ⟷ very bad
2 ⟷ bad
3 ⟷ satisfying
4 ⟷ good
5 ⟷ very good

Since our goal is to determine the reviews polarity, we keep only two labels using the following approach:

- Label 1: if the rating is equal to 4 or 5
- Label 0: if the rating is equal to 1 or 2

The reviews with rating 3 are dropped because, sometimes they carry both positive aspects, and sometimes they are more to be negative, so it cannot be attributed to a specific class, it depends on the user. The total size of the dataset is 520 000 samples.

As for the scraped data, it is related to two products categories:

- Socks data: 1479 reviews with ratings between 1 and 5
- Makeup Data: 712 reviews with ratings between 1 and 5

The data is labeled similarly to Amazon Review Dataset (binary labels: 0 and 1).

III.3.2 Experimental Protocol

We test SVM and LSTM in our sentiment analysis system for product's forecast improvement, since we use classification models, we compute the accuracy, precision, recall and F1 score.

Accuracy is the most intuitive performance measure and it is simply a ratio of correctly predicted observation to the total observations:

$$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions} \quad \text{III.1}$$

The classification accuracy can be misleading, for example, in a problem where there is a large class imbalance (more samples in one class than the other), a model can predict the value of the majority class for all predictions and achieve a high classification accuracy, precision recall and F1 score are used to evaluate the performance on both data classes.

Precision is the ratio of correctly predicted positive observations to the total predicted positive observations. The question that this metric answer is of all reviews that were labeled as positive, how many actually are positive?

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \quad \text{III.2}$$

Where, True Positives (TP) represent a positive instance is classified as positive, if it is classified as negative, it is counted as False Positives (FP).

Recall is the ratio of correctly predicted positive observations to the all observations in actual class. The question recall answer is: of all reviews that are truly positive, how many did we label?

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negtives} \quad \text{III.3}$$

Where, True Negatives (TN) represent a negative instance is classified as negative, if it is classified as negative, it is counted as False Negatives (FN).

F1 Score is the harmonic mean between precision and recall:

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall} \quad \text{III.4}$$

This metric takes in consideration both the precision and recall of the model and translates them into one numerical value.

To evaluate the forecast with and without the sentiment intensity feature for online products we use the Root Mean Square Error (RMSE) It is the standard deviation of the residuals (prediction errors). Residuals are a measure of how far from the regression line data points are, RMSE is a measure of how spread out these residuals are. In other words, it tells you how concentrated the data is around the line of best fit.

The corresponding formula is the following:

$$RMSE = \sqrt{\frac{\sum_1^N (Predicted_i - Actual_i)}{N}} \quad \text{III.5}$$

Finally, we will be testing SVM and LSTM execution time on Amazon Reviews Dataset to measure their performance:

- For training SVM it takes about 24 hours while LSTM takes 8 hours.
- After training the models will be saved for further usages, as a result testing will be time negligent compared to training, in fact, for SVM it is 30 seconds while for LSTM it takes 15 seconds.

II.3.3 Sentiment Analysis Results

We have two testing use cases for each of our systems: for the first one we work on online reviews and the second one we work on tweets.

III.3.3.1 Sentiment Analysis System for Demand Forecast

After the two steps of Data cleaning and Text vectorization, presented in Chapter II section II.2, we apply the chosen models in step Model selection, presented in Chapter II section II.2, on Amazon Reviews Dataset, and the final output is a sentiment intensity for each sample text, which we take as the probability of the predicted class.

The data is split into 80% for training and 20% for validation, to better evaluate the performance of our models we do multiple training runs while shuffling the data at each iteration, calculating training and validation accuracy and comparing error metrics.
Finally, we use the scraped data from Infor's client official website as test data for our models.
The models that we are going to test are:

- **TF-IDF + SVM:** SVM requires a vector representation of the training sample (a sentence in our case) while Word Embedding gives a feature vector for each word, thus we need to use Word Embeddings with that considers the order of different feature values (the order of the words in our case). TF-IDF returns a numerical representation of the sentence while highlighting important words, it can be used with SVM model since it works with any type of numerical vectors.

- **Word Embeddings + LSTM:** LSTM is a sequential model, it considers each word as an input in the total sequence, in fact, it uses each word vector representation as an input to its neighbor, as it forward propagates its calculated values, TF-IDF will not in this case since it returns a representation of the whole sentence. Using those two together means: using Word Embeddings first to learn the contextual representations of words, examine, in second, the order of the different terms, so that we determine the total sentiment of the sentence.

A. Hyperparameters Tuning

A hyperparameter is a parameter whose value is set before the learning process begins. By contrast, the values of other parameters are derived via training. Given these hyperparameters, the training algorithm learns the parameters from the data.

Tuning the hyperparameters consists in finding the best set of parameters for our model, this process can be intuitive by choosing the best set of parameters based on field knowledge, or the other alternative is to do multiple training runs with different parameter combinations and pick the set that returns the best metrical results.

Tuning TF-IDF + SVM:

For TF-IDF, we use an n-gram range of [1,2], this means that our features can be constituted of one or two contiguous words. In fact the more we increase the n-range, the more we consider word's far neighbors. So, the size of our feature vector increases exponentially, and the training process becomes more difficult. The chosen value makes a compromise between modeling context and training time.

For SVM, we focus on tuning the regularization parameter C in order to avoid overfitting the data. Figure III.2 shows different values for train and test accuracy with increasing C parameter within a defined range.

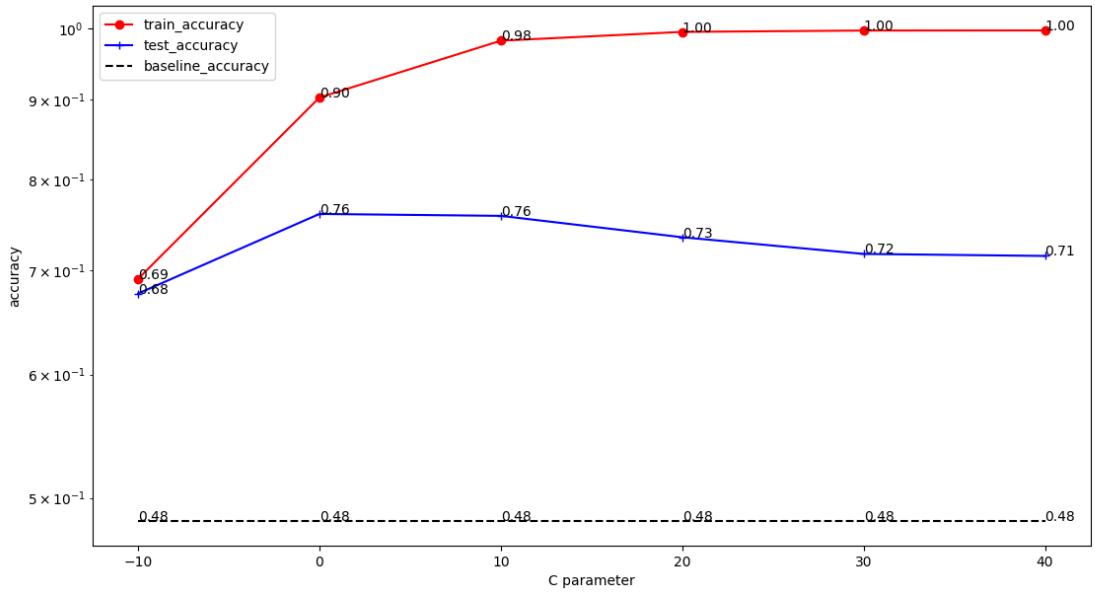


Figure III.2: Evolution of accuracy in function of regularization parameter for TF-IDF+SVM model.

The C parameter represents the inverse of the regularization parameter, we expect the training accuracy to increase with C values with a drop in the validation accuracy. The chart shows similar results, in our case, the second point indicates the best C value.

Tuning Word Embeddings + LSTM:

For word embeddings we have the possibility to choose between Continuous Bag of words or Skip-gram model (presented in chapter II, section II.2.3), we used the first solution since we wanted one representation of each word with regard to its overall context (Skip-gram model returns multiple representations of a word in function of its different contexts). We choose window size of 3, this value is used to define the word's neighbors used for context capturing.

Finally, after doing multiple training runs, the size of the vector used for representing words was set for 100. This value is a compromise between riche representation of the word and long training time.
For LSTM, the tuned hyperparameters are:

-Number of epochs

-Number of units

-Batch size

-Dropout

-Recurrent dropout

Python tool **Tensorboard** was used to visualize the evolution of validation accuracy regarding the different hyperparameters, Figure III.3 shows the evolution of validation accuracy in function of the number of epochs.

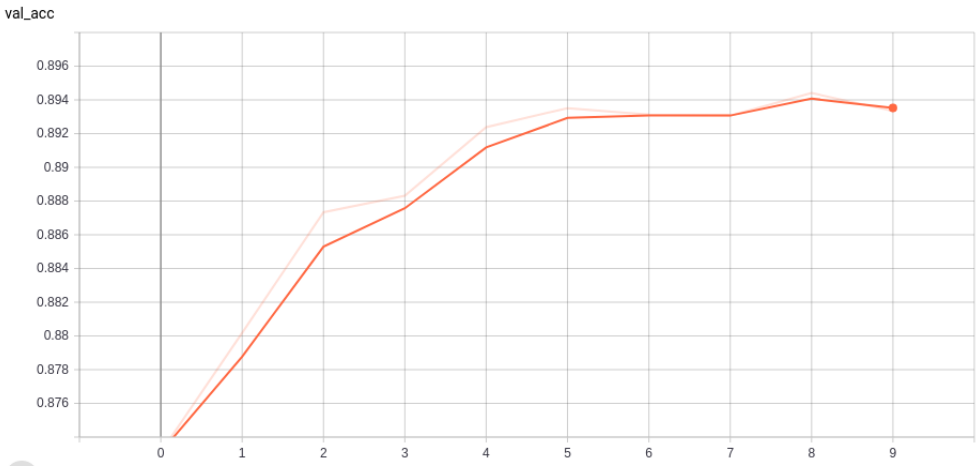


Figure III.3: Evolution of model validation accuracy in function of number of epochs.

We determine here the best number of epochs that gives us the highest accuracy.

The other parameters are tuned in similar manner by running multiple iterations and picking the best set.

B. Results and Discussion

Table III.1 resumes the performance of the two models regarding the defined metrics.

Table III.1: Metrics comparison between models.

	Train Accuracy	**Validation Accuracy**	**Precision**	**Recall**	**F1 score**
TF-IDF+ SVM	0.90	0.76	0.79	0.74	0.76
Word embeddings + LSTM	0.92	0.89	0.89	0.90	0.89

The table shows that Word embeddings + LSTM gives better results in term of accuracy, precision, recall and F1 score, there is gap of 15% in performance between the two models. This is expected, because LSTM has the capacity to differentiate between different words orders, he can detect the global meaning of the sentence.

SVM's limit is its incapacity to model word's context in the sentence. TF-IDF with n-gram features can be used to model neighboring words, though, it cannot compete with LSTM large memory cell.

We measure the performance of the models on the scraped data as shown in Table III.2.

Table III.: Model's comparison on two different product's data.

	Test accuracy (Socks Data)	**Test accuracy (Makeup Data)**
TF-IDF+ SVM	62.5%	61.9%
Word embeddings + LSTM	91.2%	89.4%

Word embeddings + LSTM gives better tsting accuracy than the TF-IDF + SVM on both datasets (socks data and makeup data). Since the test is run on two different product's categories, the results show the capability of the Word embeddings + LSTM model to generalize what it has learned during the training process, this proves that our sentiment analysis model can classify data from different backgrounds.

C. Impact on the Forecast

We use LSTM to estimate the sentiment intensity of the scraped reviews, since It has shown better performance than SVM, and then pass it as a feature to Factorization Machine, (presented in chapter II, section II.2.5), for Forecast measurement.

Table III.3 compares the RMSE value of the forecast in percentage with and without the sentiment feature.

Table III.3: RMSE value with and without the sentiment feature.

RMSE without sentiment feature	RMSE with sentiment feature
73.23%	72.6%

The RMSE value decreases of 1% using the sentiment feature, there is a slight improvement, but it does not meet what we were expecting, in real life, customer's opinion does have an impact on product's volume of sales, if we do not see this impact on the forecast accuracy, it means that there is a problem in modeling this phenomenon, in fact we come up with three different hypothesis:

First, the review's data on retailer's website does not reflect what all customers are feeling, maybe only a small fraction on the target consumer's express themselves or consult the website when making the purchase decision, the solution is to look for another source of data that is likely to reflect the overall customer's attitude.

Second, if nothing is wrong with our data, then, the sentiment analysis model (LSTM) could not extract the exact sentiment information from text, this hypothesis is rejected since the model has high test accuracy on different products categories.

Finally, the customer's attitude towards products is likely to change through time, therefore, we should introduce a temporal variation to the product's sentiment intensities, before passing them to the forecasting model.

III.3.3.2 Sentiment Analysis System for Brand Perception

A. Approach

After going through the cleaning step, (presented in chapter II, section II.2.2z, the tweets are passed to the VADER model which gives three different numerical values:

- Neutral sentiment intensity
- Positive sentiment intensity
- Negative sentiment intensity

The different values range between –1 and 1with 0 as mean value, to estimate the sentiment intensity for each tweet, we go through the following steps:

Filter neutral tweets: We are interested here in the returned neutral sentiment intensity, the goal is to filter all the objective tweets and keep only those that carry some subjectivity in them. Filtering the tweets based on VADER's value and with threshold of 0 does not give very satisfying results, therefore, we go through a manual process of reading different samples and analyzing their returned sentiment intensity, this allows us to determine the adequate threshold for better sentiment analysis results. For a value between -1 and 1, all the tweets with sentiment intensity higher than 0.7 are considered neutral.

Filter negative tweets: The approach is similar previous step except that our interest here is the sentiment intensity value, the goal is to determine from the remaining tweets those that express a dissatisfaction of the user.

Filter positive tweets: Once we eliminate the neutral and negative tweets, what remains is considered as positive, the reason why we have left positive reviews to the end, because even negative reviews contain some positivity in them, therefore, if we inverse the order of the last two steps, we may fall into the trap of considering all the reviews as positive.

Aggregate sentiment intensities: we take the average of all the sentiment intensities for each day, then we aggregate the results per week to obtain our final output.

B. Results and Discussion

The extracted value from the collected tweets is a sentiment intensity that measures the customers degree of satisfaction and which varies per week, this is shown in Figure III.4 where we are analyzing customers attitude towards one of Infor's retail clients.

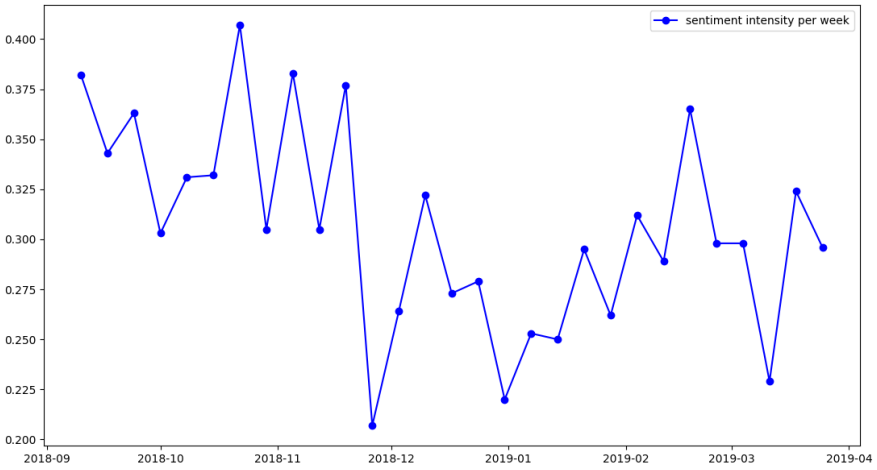


Figure III.4: Evolution of sentiment intensity per week.

The calculated sentiment intensity cannot be used as feature for products sales forecasting similar to what we have seen for online reviews, because, this value evolves through time and so it is not known until we reach it in the present, as a result it cannot be used to predict a value in the future.

However, the previous chart is useful for giving the retailer an idea about the evolution of his customers degree of satisfaction, in fact, he would have an overview about the particular dates that affected his brand reputation, and based on this, he may try to investigate what went wrong and then take the adequate measures for problem troubleshooting.

The lowest drop in sentiment intensity coincides with first of December, this week contains the Black Friday event in USA, where customers get promotions on different products. This drop is explained by the dissatisfaction of customers towards the offered services and promotions. Thus, the company needs to detect what made the customer's angry, and learn for future events.

General Conclusion

Customer's feedback analysis constitutes a major factor for retail business prosperity. In fact, how a company sees its products is different from what the consumer really perceives, this urges retailers to find methods for mining their client's opinions and predicting their future demand. In this project we focused on sentiment analysis applications in retail industry, in one hand, we built a system that scrapes product's related reviews from Internet, cleans the data, and then determines the customers opinion polarity, we compared SVM versus LSTM model but the latter gave us far better results, in fact, LSTM has 90% for validation accuracy, his performance was tested and validated on reviews with different contexts. The model also returns a measure of customer's satisfaction and dissatisfaction in the form of a numerical value, it is then passed as a feature to Infor's existing solution for sales forecasting, the aim is to have a better estimation of customer's future demand for reducing unnecessary costs in other retail services, although, the impact of sentiment feature was not really significant on the forecast, a potential improvement is to add a temporal variation to the extracted value, in fact, we are generating a sentiment value for each product, we then use it for his sales forecasting, so, in other words, we suppose that customer's opinion will not change through time, this is likely not the case in real life examples, and thus, our system could be improved through extracting a sentiment intensity for products that varies through time.

In the other hand, we build a system that gives retailers an idea of their customer's feedback and how it evolves through time, the goal is to give insight about what is happening on the consumer's side, we work on tweets and extract how people are feeling about a specific brand, the customer's opinion is determined using a group of sentiment lexicons and grammatical rules. Finally, we give retailer an overview of his customer's feedback through a chart that displays the evolution of sentiment intensity through time, this latter is helpful for analyzing customer's degree of satisfaction, understanding the history and what went wrong at specific point of time, so we take better decisions in the future.

Whenever there is a drop in the sentiment intensity chart, it means there a problem on customer's side, an extension of this solution consists of detecting automatically the exact problem, and showing to the retailer details about what went wrong.

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